

Galton's Bend: An Undiscovered Nonlinearity in Galton's Family Stature Regression Data and a Likely Explanation Based on Pearson and Lee's Stature Data

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Abstract

In Francis Galton's 1886 paper "Regression Towards Mediocrity in Hereditary Stature," Galton analyzed the heights of 928 adult children and their 205 pairs of parents to illustrate his linear regression model. Although Galton's data have been recalled frequently to illustrate linear regression and regression toward the mean, no one seems to have noticed that his height data do not fit his model. The purpose of this paper is both to reveal this curiosity and to find a possible explanation for its existence using related data from Galton's colleague Karl Pearson

1 Introduction

Francis Galton devised his regression model to develop an evolutionary theory of heredity. As Stigler (1986) shows, the mathematics of linear least squares fitting date back at least to the early 19th century. But it was Galton's idea of regression based on the bivariate normal distribution that allowed the development of coefficients of heredity supporting a theory of natural inheritance. Galton derived his theory by looking at data, but the lens he used profoundly shaped what he saw.

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2 Galton's Analysis

Figure 1 contains a graph from Plate X of Galton (1886). The data underlying this graph are found in Table I of Galton's paper and reproduced in Stigler (1986, p. 286) and Stigler (1999, p. 181). Galton's table contains tallies of the height of 928 adult children grouped by the average height of their parents. In this table, Galton adjusted the heights of female children to correspond to the male heights by multiplying them by 1.08. As he says,

In every case I transmuted the female statures to their corresponding male equivalents and used them in their transmuted form, so that no objection grounded on the sexual difference of stature need be raised when I speak of averages. The factor I used was 1.08, which is equivalent to adding a little less than one-twelfth to each female height. It differs a very little from the factors employed by other anthropologists, who, moreover, differ a trifle between themselves. (Galton 1886, p. 247)

Galton describes how he arrived at the graph from the data in his table.

I found it hard at first to catch the full significance of the entries in the table, which had curious relations that were very interesting to investigate. They came out distinctly when I "smoothed" the entries by writing at each intersection of a horizontal column with a vertical one, the sum of the entries in the four adjacent squares, and using these to work upon. I then noticed (see Plate X) that lines drawn through entries of the same value formed a series of concentric and similar ellipses. Their common centre lay at the intersection of the vertical and horizontal lines, that corresponded to $68\frac{1}{4}$ inches. Their axes were similarly inclined. The points where each ellipse in succession was touched by a horizontal tangent, lay in a straight line inclined to the vertical in the ratio of $\frac{2}{3}$; those where they were touched by a vertical tangent lay in a straight line inclined to the horizontal in the ration [*sic*] of $\frac{1}{3}$. These ratios confirm the values of average regression already obtained by a different method, of $\frac{2}{3}$ from mid-parent to offspring, and of $\frac{1}{3}$ from offspring to mid-parent, because it will be obvious on studying Plate X that the point where each horizontal line in succession is touched by an ellipse, the greatest value in that line must occur at the point of contact. The same is true in respect to the vertical lines. These and other relations were evidently a subject for mathematical analysis and verification. (Galton 1886, pp. 254–255).

Galton goes on to describe how he consulted with the mathematician J. Hamilton Dickson at Cambridge University to derive the equations for this ellipse from the bivariate normal distribution and to compute from the normal model the exact estimates for the slopes of the lines in the figure.

2.1 An Adaptive Fit to Galton's Data

Figure 2 contains a SYSTAT rendering of Galton's figure. The approximately 68 percent confidence ellipse is sized to match Galton's ellipse based on one probable error (Galton's "probable deviation"). The symbols in the figure have been jittered with a small amount of random error to highlight the density. Galton's lines have been colored light gray.

The dark curve in the center of the plot is a *loess* smoother (Cleveland and Devlin, 1988). The smoother suggests that the relation between parent and child stature is not linear. There is a bend in the curve somewhere around the average height of approximately 68 inches for parents and children. A two-stage piecewise linear regression (Hinkley, 1971) identifies a breakpoint at around 70 and finds it highly significant ($p < .0001$).

If Galton's data are fit better by a piecewise linear model than by a simple linear model, what could be the cause? One possibility is that Galton pooled and aggregated over disparate populations. We need to separate fathers, mothers, sons, and daughters.

3 Pearson's data

Galton's disciple Karl Pearson had access to Galton's height data and analyzed them in Pearson (1896) and in Pearson and Lee (1896). Unfortunately, Pearson's papers do not show Galton's data separated by sex. Pearson and Alice Lee did collect a similar set of height data from English families during roughly the same time period, however. Pearson and Lee (1903) contains cross tables of father and son, father and daughter, mother and son, and mother and daughter heights from this more extensive dataset.

Figure 3 shows the full gender cross-tabulation of Pearson and Lee's data. We have superimposed confidence ellipses and *loess* smoothers in each cell. The two bottom panes show *loess* regressions of mother heights on son and daughter heights. The bend appears in the smoothers in both lower panes in the figure.

3.1 Reproducing Galton's result from Pearson's data

Can we pool Pearson's data and get Galton's result? Figure 4 shows child heights plotted against parent heights when daughter and mother heights have been multiplied by Galton's adjustment factor of 1.08. With these blended data, the telltale bend at the lower end of the fitted *loess* regression is readily apparent.

Finally, if we fit a piecewise linear model to the blended Pearson data, we get a significant breakpoint near the mean child height of 65 inches ($p < .05$).

4 Conclusion

Is there any chance that Galton and Pearson, with the statistical tools they had available, could have discovered this anomaly? Galton smoothed his data with a two-dimensional rectangular counting kernel, the “naive method” described by Silverman (1986), in order to regularize the ellipse he sought. But it may be unreasonable to assume that Galton might have used a conditional local smoother to assess the fit of his regression line and, further, to recognize whether a bend in the smoother was important. In their search for universal hereditary laws, Galton and Pearson were driven by the linear model and the normal distribution because the associated parameters had scientific meaning for them that went beyond mere description.

Could Galton and Pearson have used their *linear* tools to detect an anomaly and avoid pooling? In the context of Galton’s linear regression model, we might ask if the mother data support a different regression slope from that of the father data. Applying a simple 2x2 analysis of covariance, with child height as the covariate, we find the test for homogeneity of slopes with respect to parent (mother/father) to be significant ($p < .01$), while the same test with respect to child (son/daughter) is not significant. Although we should qualify our conclusions because of within-family dependence in the observations, we find scant support for pooling these data.

Galton was clearly sensitive to the problem of pooling data from disparate groups. He wrote, for example,

It clearly would not be proper to combine the heights of men belonging to two dissimilar races, in the expectation that the compound results would be governed by the same constants. (Galton 1869, p. 29, cited in Stigler, 1986)

Galton was also sensitive to how gender differences in stature might affect his conclusions.

I use the word parent to save any complication due to a fact apparently brought out by these inquiries, that the height of the children of both sexes, but especially that of the daughters, takes after the height of the father more than it does after that of the mother. My present data are insufficient to enable me to speak with any confidence on this point, much less to determine the ratio satisfactorily (Galton 1886, p. 250).

Pearson pursued this point further. Analyzing Galton’s data, he found that the father–son correlation was .40, the father–daughter was .36, the mother–son was .30 and the mother–daughter was .28 (Pearson, 1896). Pearson viewed these correlations as consistent with Galton’s interpretation.

...both sons and daughters, on the average, take very considerably more after their father than after their mother (Pearson, 1896, p. 275).

But Pearson distrusted Galton's correlations because of the informal nature of Galton's sample. Pearson examined the corresponding correlations in the data he and Alice Lee had collected. Pearson found all four of his correlations to be approximately .50 (Pearson and Lee, 1903). From this failure to find differences among correlations in his own data, Pearson later concluded that the asymmetry in Galton's data was due to mis-measurement:

I think it may well have been due to amateur measuring of stature in women, when high heels and superincumbent chignons were in vogue; it will be noted that the intensity of heredity decreases as more female measurements are introduced. Daughters would be more ready to take off their boots and lower their hair knots, than grave Victorian matrons (Pearson, 1930, page 18).

Although Pearson noted the almost equal correlations in his and Lee's data, he did not emphasize the differences in standard deviations. In another context, Pearson noted that the standard deviations did differ

Comparing the standard deviations of fathers and sons, we see that fathers and sons are within the limits of random sampling equally variable. On the other hand daughters' standard deviations are in every case sensibly larger than those of their mothers (Pearson and Lee, 1903, page 371).

But Pearson did not note that **the significantly lower standard deviations for the mothers might lead to nonhomogeneity of slopes among the four groups even after multiplying the women's data by 1.08**. As we have seen, a test for homogeneity of slopes on Pearson and Lee's data fails to support pooling.

This is not the first, nor likely the last, example in which improper pooling can lead to mis-specification. It is interesting, nevertheless, that one of the most famous datasets in the history statistics has kept its secret for so long. The secret survived Pearson's close scrutiny perhaps because he, like Galton, was determined to pool the data in order to compute general heredity coefficients. It also escaped the attention of those citing Galton's as the pre-eminent regression dataset. When we set aside Galton and Pearson's peculiar evolutionary arguments for a pervasive normal law of heredity, we are better able to see disparity in the actual numbers. Prior expectations influence posterior judgments.

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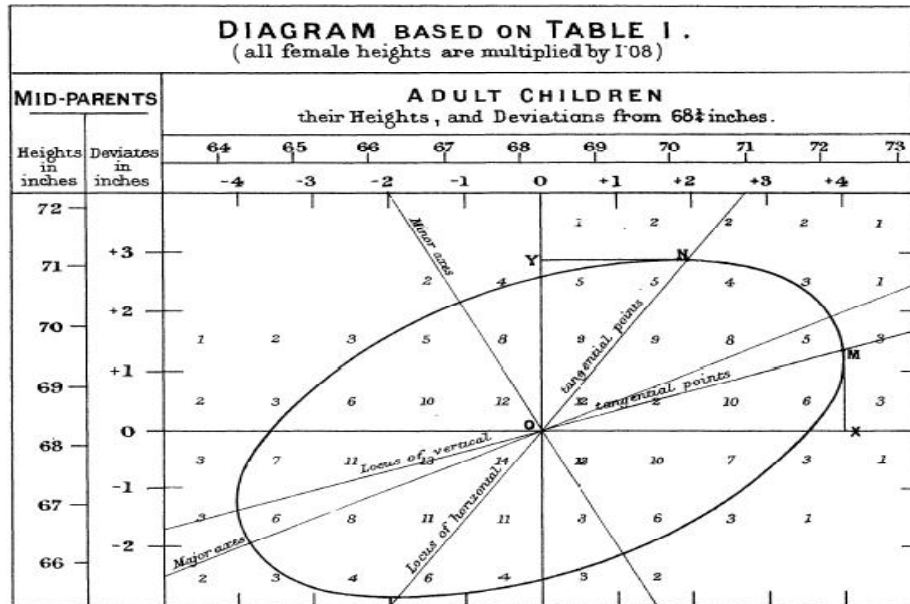


Figure 1: Galton's Fitted Regression Model

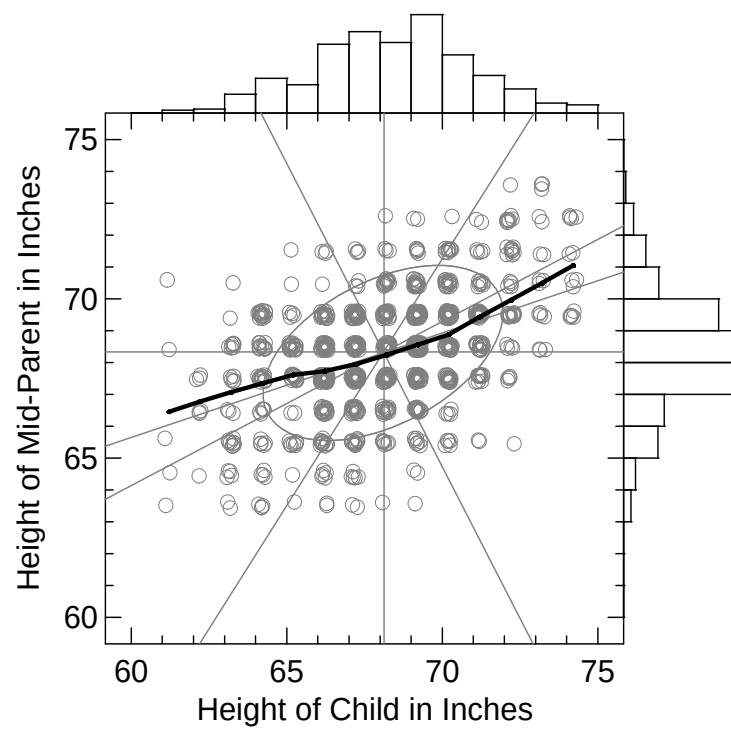


Figure 2: SYSTAT plot of Galton's Data with *loess* fit

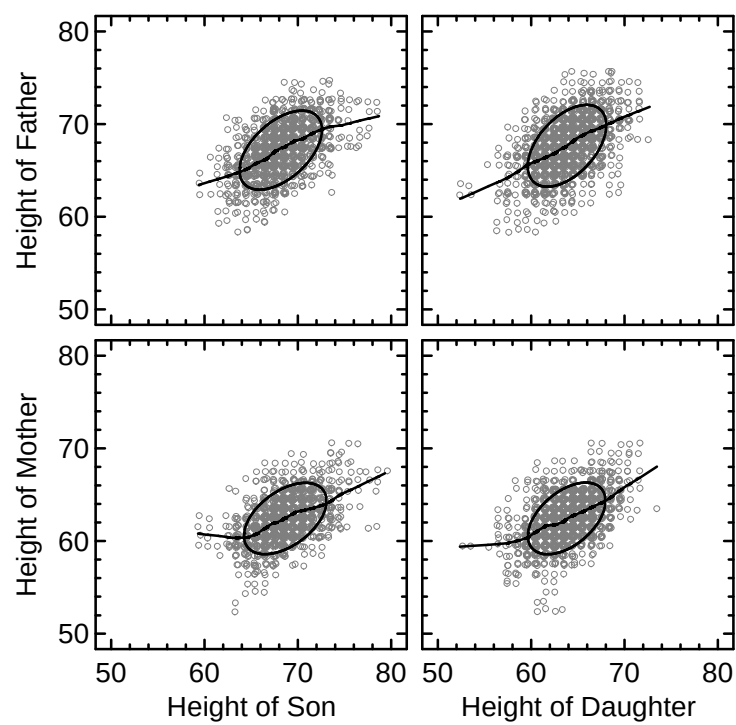


Figure 3: Pearson's Data

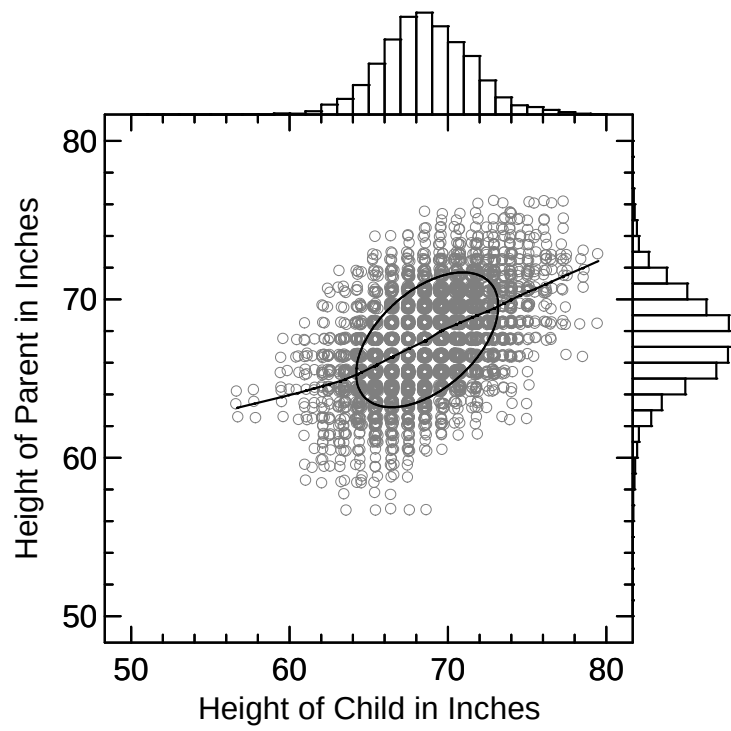


Figure 4: Reproducing Galton's Result from Pearson's Data